Volume III · Chapter 14 — Training Pipeline: Pre-training → Fine-tuning → Alignment (DPO) · Labs

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-III-Chapter-14-context.md` for the AI interaction log.

Prerequisites: Ch 11, Ch 12.

Learning outcomes — the student can: describe the full three-stage pipeline; reason about data/compute for pretraining; run domain fine-tuning; build preference data and apply DPO; compare RLHF vs. DPO; manage training runs with tracking.

Labs (hands-on) — 20 h:

#	Lab	h
14a	Continued pretraining (small) on a biomedical corpus	5
14b	Build an SFT dataset & run supervised fine-tuning	5
14c	Create a preference dataset (chosen/rejected) for a clinical behavior	3
14d	Run DPO on the SFT model	5
14e	Track experiments & compare stages (metrics, safety checks)	2

Datasets/tools: HF `transformers`/`trl`/`peft`, DPO; experiment tracking (W&B or MLflow); GPU. **Assessment:** aligned model across stages (**60%**); training report (**20%**); quiz (**20%**). **Key decisions:** pretrain vs. fine-tune vs. align scope; DPO vs. RLHF; data quality vs. quantity; compute budget. **References:** plan §13 → *Fine-tuning, alignment & benchmarks*; *Medical LLMs*; *Clinical datasets*. **Hours:** Theory **16** + Lab **20** = **36**.